

2078

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Graphics**

Code No: **CSC214** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **rendering over offline rendering? Discuss the limitation of Z-Buffer algorithm.11.Describe the requirement for line clipping. Explain the scan line polygon filling algorithm.12.What is quadric surface? Compare between diffuse reflection and specular reflection.**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define orthographic, parallel and perspective projections. Consider a region defined by the position vector $P = 1121212122211221$, relative to global XYZ axis system. It is rotated by $+30^\circ$ about X-axis and passing through point (1.5, 1.5, 1.5). Find the final position of the region.
2. What is the method to recognize boundary point and interior point in solid modeling? Describe how BSP recursively subdivides a space into convex sets.
3. List some significances of virtual reality. Differentiate between virtual reality and augmented reality with example. Demonstrate how a polygon can be created using OpenGL.

Section B

Attempt any Eight questions[8×5=40]

4. Discuss the strength and weakness of the human visual system. Describe Spline representation for the curve.
5. Plot the 1st octant of a circle centered at origin, having the radius 10 units.
6. Define fractal. Explain the Bezier curve and B-Spline curve.
7. Find the new co – ordinate of the triangle ABC, with co-ordinates A(0, 0), B(1, 1) and C(5, 2) after it has been magnified to twice of its size.
8. What is the task of polygon table? Why we have to remove hidden surface? Explain with any one methodology?
9. Define intensity attenuation. Distinguish between Gouraud shading and Phong shading model.
10. What is the advantage of real time rendering over offline rendering? Discuss the limitation of Z-Buffer algorithm.
11. Describe the requirement for line clipping. Explain the scan line polygon filling algorithm.
12. What is quadric surface? Compare between diffuse reflection and specular reflection.